

High-Fidelity Representation Editing and Surgical Ablation for Cyber Ops LLMs

Scope, threat model, and safety boundary

High-fidelity representation editing—whether performed at inference-time (activation steering) or via targeted weight edits—has become a practical way to *probe, audit, and shape* the behaviors of large language models (LLMs) without full retraining. The same family of techniques is inherently dual-use: research over the last two years shows that certain safety behaviors (notably refusal) can be tightly localized in internal representation space and can be perturbed with comparatively low-rank interventions. ¹

Because your request explicitly frames an “ultimate offensive/defensive (offsec focus) cyber ops AI/LLM,” this report treats **representation engineering and ablation primarily as a defensive and assurance discipline**: (a) hardening and validating safety-critical controls, (b) measuring brittleness and tamper risk, (c) preserving core reasoning/code capabilities under distribution shift, and (d) preventing “capability drift” in agentic cyber pipelines. This emphasis is consistent with real-world signals that generative AI is already being leveraged in early-stage cyber operations by state-linked actors, raising the value of robust governance, evaluation, and system-level constraint design. ²

The security posture recommended here follows “secure-by-construction” norms for GenAI systems: threat-informed evaluation, explicit risk mapping/measurement, and controls that address both AI-specific and traditional software risks (e.g., prompt injection, insecure tool/output handling, over-agency). ³

Mechanistic foundations for behavior isolation without collateral damage

Representation engineering (RepE) is best understood as a *top-down* interpretability and control paradigm: instead of editing individual neurons, it targets **population-level representations** and operationally-defined “concept directions” to monitor/manipulate high-level behaviors. The RepE survey paper positions this as an emerging area for transparency and behavior control, spanning safety-relevant properties like honesty/harmlessness and other high-level “cognitive phenomena.” ⁴

A critical practical limitation is **causality and confounding** during representation “reading.” CARE (CAusal Representation Engineering) argues that RepE often assumes models consistently follow the instructed role during activation collection; violations introduce confounded correlations that can make interventions unreliable. CARE’s matched-pair trial design is proposed specifically to control confounders and make the activation–behavior link more causal rather than merely correlational. ⁵

Within this landscape, refusal has become a canonical case study. The NeurIPS work “Refusal in Language Models Is Mediated by a Single Direction” reports that, across many open chat models, refusal can be associated with a low-dimensional (often effectively one-dimensional) subspace; perturbations along that

internal direction can dramatically alter refusal behavior. ⁶ However, follow-on work (AAAI 2026) cautions that “single-direction” views can be incomplete: refusal may occupy a *low-dimensional manifold*, and extracting multiple directions via Self-Organizing Maps can outperform single-direction suppression in their experiments. ⁷

The central engineering challenge you highlighted (cross-distribution contamination, capability degradation, collateral feature loss) is strongly supported by empirical evidence: **naïve ablation of a “dirty” contrastive direction** can induce substantial distribution drift and capability-tax effects. Surgical Refusal Ablation (SRA) frames this as *polysemantic entanglement*—the raw contrastive vector often mixes refusal signal with syntax/style and core capability circuits (coding/math/reasoning)—and reports large drift under standard ablation versus much smaller drift after “concept-guided” cleaning. ⁸ A broader cross-architecture comparison paper similarly finds that “ablation” outcomes can vary widely by tool/model, with mathematical reasoning often among the most sensitive axes—again underscoring that **capability preservation cannot be assumed** and must be measured. ⁹

Iterative hard-negative refinement as a precision and robustness amplifier

Your ARCHITECT note to “integrate iterative hard-negative refinement” aligns with one of the key operational lessons emerging from the most successful high-fidelity protocols: static prompt suites under-cover the rare, high-leverage failure modes that actually matter in deployment (especially for cyber ops agent workflows, where edge cases are the norm).

SRA explicitly incorporates an **iterative hard-negative refinement loop**: prompts that still trigger the undesired behavior after an intervention are fed back into the estimation procedure to refine the underlying direction used for subsequent cleaning/intervention rounds. ¹⁰ Conceptually, this acts like an *active learning* layer on top of representation reading: the system continuously re-focuses the learned direction toward the most robust expression of the target behavior, rather than overfitting to a convenient “template ensemble.”

For cyber ops models, this idea generalizes beyond refusal. Any operationally-defined behavior you care about (e.g., “tool-call only with explicit authorization,” “no cred exfiltration,” “no unsafe shell primitives,” “tight JSON schema adherence,” “no hallucinated CVEs”) will have hard-negative tails. Hard-negative refinement should therefore be treated as a **standing assurance loop**: (1) mine failures from realistic agent traces, (2) update the evaluation stimulus set, and (3) re-measure both safety outcomes and capability preservation. This is the most direct way to prevent “quiet regressions” that only surface under adversarial or multi-step tool-use conditions. ¹¹

Probabilistic ablation parameters and stochastic scheduling for robust safety

Your second ARCHITECT note—introducing stochastic scheduling / probabilistic ablation parameters—maps closely onto a major defensive trend: **training models to remain safe even when internal refusal features are partially suppressed**.

DeepRefusal (EMNLP Findings 2025) is a representative example. It motivates an “alignment depth problem” (over-reliance on the first tokens) and argues that jailbreaks can inhibit refusal-related activations; it then proposes probabilistically ablating the refusal direction across layers and token depths during fine-tuning, forcing the model to *rebuild robust refusal mechanisms* even under internally adversarial conditions. The paper reports substantial reductions in jailbreak attack success rates (on the order of ~95% in their setting) while maintaining capabilities with only modest degradation. ¹²

From a cyber ops perspective, the key transferable idea is not “probabilistic edits” per se; it is **stochastic robustness training against internal mechanism disruption**. In security terms, DeepRefusal treats “refusal-direction manipulation” as an internal fault injection threat model and trains invariances against it. ¹³ This is especially relevant if your deployment assumes any combination of (a) open-weight models, (b) partially untrusted runtime environments, (c) extensible tool ecosystems, or (d) high-value targets where adversaries may attempt tampering and control-plane compromise.

A second, independent argument for probabilistic evaluation and intervention comes from research on steering reliability. The NeurIPS 2024 steering vectors paper finds that steerability can be highly variable across inputs, with spurious biases affecting effectiveness; this implies that **single-shot, deterministic intervention evaluation can materially mis-estimate risk** unless uncertainty and variability are explicitly measured. ¹⁴

Stability guards for complex architectures and MoE-weight integrity

Your third ARCHITECT note—Weight-SVD stability guards for Mixture-of-Experts (MoE)—is consistent with the fact that “behavioral circuits” are being investigated in increasingly heterogeneous architectures.

On the model side, OpenAI ¹⁵’s gpt-oss family is explicitly MoE: the public release describes gpt-oss-120b as a 36-layer Transformer with MoE routing (128 experts with 4 active per token), and reports 117B total parameters with ~5.1B active parameters per token (plus long-context support up to 128k). ¹⁶ These architectural facts matter for representation editing because (a) expert routing creates conditional computation pathways, and (b) “where” and “how” behavior is represented may differ from dense-only models, increasing the risk of interaction effects between edits and routing dynamics. ¹⁶

The most direct research link to your requested “SVD stability guard” is the 2026 arXiv preprint “Universal Refusal Circuits Across LLMs.” It frames refusal transfer as a *measurement tool* for universality (not a deployment recommendation) and introduces a “weight-SVD stability guard” designed to project interventions away from high-variance weight subspaces to reduce collateral capability damage. ¹⁷ The paper explicitly evaluates across multiple donor/target pairs including Dense→MoE settings and emphasizes that the guard is a key ingredient for avoiding catastrophic degradation. ¹⁸

For “ultimate cyber ops AI” engineering, two defensive implications follow.

First, **tamper resistance becomes a first-class requirement**. If sophisticated behavior edits are feasible with low-rank interventions, then open-weight and locally-deployed models must be treated as high-value assets: weight integrity, provenance, and controlled update channels are security controls, not

afterthoughts. This aligns with OWASP's focus on supply chain vulnerabilities and model theft as top risks for LLM applications. ¹⁹

Second, the same mathematical intuition behind stability guards can be repurposed for **safe capability-preserving alignment transfers** (e.g., transferring protective “constraint behaviors” into a newly trained cyber-analyst model without sacrificing its vulnerability triage accuracy). The universality work explicitly positions its method as a universality test and includes “knob/budget audits” to reduce hidden tuning degrees of freedom—an approach that generalizes well to regulated or high-assurance cybersecurity environments. ²⁰

Verification protocol with statistical rigor for cyber ops deployment

A central theme across the literature is that “no measurable degradation” is only meaningful if you specify **what you measured, how sensitive the measurements are, and what uncertainty remains**.

Benchmark Profiling and the Ability Impact Score (AIS) provide one of the more mechanistically-motivated approaches to breaking apart “aggregate benchmark performance.” The Benchmark Profiling paper argues that benchmark labels can mislead (benchmarks often draw on multiple abilities), and it computes AIS by combining gradient-based importance scoring with targeted parameter ablation to quantify each ability's contribution to benchmark performance. ²¹ This is directly compatible with your “AIS mapping” requirement, and also supports your constraint that deductive logic, math reasoning, and code synthesis should be preserved: AIS can be used as a *capability profile regression test*, not just an overall-score snapshot. ²¹

However, the statistical point from your ARCHITECT note is essential: uncertainty must be quantified. Two different sources reinforce why. First, steering vectors can have high variance across inputs and can be influenced by spurious prompt-template factors, meaning point estimates can be misleading without error bars. ¹⁴ Second, mechanistic evaluation tools themselves have pitfalls: activation patching is widely used, but “how to use and interpret activation patching” stresses subtleties in application and interpretation, including metric choice and failure modes—again indicating that a *single metric, single run* is rarely adequate for high-stakes conclusions. ²²

A defensible cyber-ops-grade validation stack therefore looks like this (expressed as an assurance pipeline rather than an implementation recipe):

Capability preservation (primary constraint).

Use a mix of (a) ability-profile regressions (AIS profiles across a stable benchmark suite) ²¹ and (b) high-resolution distributional proxies that can detect “soft damage” earlier than discrete accuracy. SRA reports exactly this motivation: it uses teacher-forced perplexity on targeted domains (e.g., math/code) as a sensitive proxy for capability distribution disruption, while also reporting distribution drift via (first-token) KL divergence and perplexity on a standard corpus. ²³

Distribution drift and locality.

Continue to treat KL divergence and perplexity shifts as *minimum necessary* checks (they catch large distribution shifts), but do not treat them as sufficient: the steering reliability literature shows that steering can fail or reverse on subsets of prompts even when average metrics look acceptable. ¹⁴

Ablation-method hygiene for interpretability measurements.

For exploratory mechanistic studies (pre-edit diagnosis), replace “zeroing” assumptions with better-matched ablation constants when possible. Peak Activation Centering (peak ablation) is proposed precisely because different ablation constant choices can induce different performance degradations, and resampling is often highly disruptive. ²⁴ Optimal Ablation (OA) similarly argues that the value used to replace an ablated component matters, and proposes an “optimal” ablation procedure with theoretical and empirical advantages over common ablation methods. ²⁵

Cyber-specific evaluation gates (defensive).

For a cyber ops model, you should validate on *defensive* and *assurance* tasks, not just generic language benchmarks. Repository-level vulnerability detection benchmarks (e.g., JITVul) are designed to evaluate more realistic interprocedural vulnerability detection, and report that agentic methods can outperform raw prompting in distinguishing vulnerable vs. benign code—useful as a measure of whether your edits harmed practical security analysis skill. ²⁶ Separately, work on “CVE-Bench” style environments highlights that LLM agents can be evaluated in sandboxed settings for offensive capability—important for *risk assessment* and red-team containment design—but the key engineering move is to use these to measure and minimize misuse pathways, not to enhance them. ²⁷

System-level threat controls for agentic cyber deployments.

OWASP’s LLM Top 10 explicitly calls out prompt injection, insecure output handling, insecure plugin design, and excessive agency as major risk categories. ¹⁹ Empirically, recent work comparing function-calling and Model Context Protocol deployment paradigms finds substantial attack success rates across many scenarios and models, reinforcing that “agent framework choice” meaningfully affects exposure. ²⁸ For high-assurance operations, model editing must be paired with platform controls and integrity gates; NIST’s AI RMF provides a lifecycle structure (govern/map/measure/manage), and NIST’s COSAIs work explicitly targets overlays for securing AI systems using SP 800-53 controls, including components like model weights and configuration settings. ²⁹

TIS // Tactical Implementation Specification (defense-first, cyber ops grade)

SITREP: Internal behavior-level interventions (RepE/ablation) can strongly affect safety behaviors and, if naively applied, can introduce capability drift and distribution shift. Refusal-related behaviors are empirically steerable and therefore must be treated as a tamper and assurance surface. ³⁰

OBJECTIVE: Deploy a cyber-operations LLM that preserves core reasoning/code/math abilities while providing robust, verifiable safety constraints and resisting jailbreak/tamper mechanisms. ³¹

BATTLE PLAN: Use causality-aware stimulus design (CARE) to reduce confounding during representation reading ⁵; incorporate hard-negative refinement loops as a standing evaluation amplifier ³²; use probabilistic robustness training paradigms (DeepRefusal-style) to harden refusal/safety circuits against internal fault injection ¹³; add stability guards (SVD-style) when transferring constraints across architectures (Dense→MoE) ³³; and enforce a multi-layer validation suite (AIS profiles + drift metrics + cyber-task regressions) with confidence intervals. ³⁴

VERIFICATION: Pass criteria must include (a) no statistically significant AIS regressions on protected abilities ³⁵, (b) bounded distribution drift under KL/PPL proxies ³⁶, and (c) stable performance on realistic defensive security tasks (e.g., repository-level vulnerability detection) ³⁷.

CONTINGENCY: If drift or AIS regressions occur, prefer inference-time or training-time robustness controls over permanent edits, and re-run matched-pair/variance-aware evaluation until error bars support equivalence (not merely “no drop observed”). ³⁸

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